

SFG 2.0,

DAY #15

(a)

In the year 2013, total no. of units sold
by company 'X' = $\frac{\text{Total sales (in ₹)}}{\text{Selling price (₹/unit)}}$

$$= \frac{3,50,000}{7}$$

$$= 50,000 \text{ units}$$

and,

In the year 2013, total no. of units sold by

$$\text{company 'Y' } = \frac{4,50,000}{5}$$

$$= 90,000 \text{ units}$$

and,

in the year 2013, total no. of units sold by

$$\text{company 'Z' } = \frac{4,00,000}{8}$$

$$= 50,000 \text{ units}$$

Total no. of units sold by all the three

U.P.S.C.

संख्या
Question No.)

इस भाग में कुछ
(Don't write any
in this part)

$$\text{Companies 'X', 'Y', and 'Z' = } 50,000 + 90,000 + 50,000 \\ = 1,90,000 \text{ units.}$$

Total Sales (in ₹) by all the three companies

$$\text{'X', 'Y' and 'Z' = } 3,50,000 + 4,50,000 + 4,00,000 \\ = 12,00,000 \text{ ₹}$$

∴ The average price of soap

$$= \frac{12,00,000}{1,90,000}$$

$$= \boxed{6.31 \text{ ₹/unit}}$$

⑤ The units sold in 2012 by company 'X'

$$= \frac{3,00,000}{6}$$

$$= 50,000 \text{ units}$$

and, the units sold in 2011 by company 'X'

$$= \frac{3,00,000}{5}$$

$$= 60,000 \text{ units.}$$

∴ The required percentage decrease

$$= \left| \frac{50,000 - 60,000}{60,000} \right| \times 100$$

$$= \frac{1}{6} \times 100$$

$$= 16.67\%$$

The percentage decrease in units sold for 'X' in 2012 over 2011 is 16.67%.

(c)

Unit sold by company 'X' in 2013 = $\frac{3,50,000}{7}$

$$= 50,000 \text{ units}$$

and, unit sold by company 'Y' in 2013

$$= \frac{4,50,000}{5} = 90,000 \text{ units}$$

and, unit sold by company 'Z' in 2013

$$= \frac{4,00,000}{8} = 50,000 \text{ units}$$

Total unit sold in 2013 by all companies = $50,000 + 90,000 + 50,000 = 1,90,000$

$$\therefore \text{The required percentage} = \frac{50,000}{1,90,000} \times 100$$

$$= 26.315\%$$

$$= 26.32\%$$

(d)

Year	No. of units sold by 'Y'
2011	$\frac{4,40,000}{4} = 1,10,000$
2012	$\frac{4,20,000}{6} = 70,000$
2013	$\frac{4,50,000}{5} = 90,000$
2014	$\frac{4,90,000}{7} = 70,000$
2015	$\frac{4,20,000}{6} = 70,000$

From the above chart, it is evident that maximum number of units sold by 'Y' is 1,10,000 in the year 2011.

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(2)

The maximum price per unit given in
the table = 9 ₹/unit

and, the minimum ^{price per} unit given in the

$$[2.00] =$$

$$= 4 ₹/unit$$

∴ The required percentage is

$$\% \text{ diff} = \frac{9 - 4}{4} \times 100\%$$

$$= \frac{5}{4} \times 100\%$$

$$= [125\%]$$